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DEPARTMENTOFINFORMATIONTECHNOLOGY

INTEGRATED MACHINE LEARNING AND REINFORCEMENT LEARNING FRAMEWORK FOR RETAIL DEMAND FORECASTING AND INVENTORY DECISION OPTIMIZATION

Abstract :

Retail businesses often face challenges in accurately forecasting product demand and maintaining optimal inventory levels. Traditional inventory management methods may result in stockouts, excess inventory, and unnecessary operational costs. To address these challenges, this project proposes an Integrated Machine Learning and Reinforcement Learning Framework for Retail Demand Forecasting and Inventory Decision Optimization. The system uses historical retail data to predict future product demand using XGBoost Machine Learning. Relevant factors such as sales history, inventory levels, pricing, promotions, seasonality, and other demand-related features are considered during forecasting. The predicted demand is then provided to a Q-Learning-based Reinforcement Learning model, which learns suitable inventory actions such as reordering, holding, or adjusting order quantities. By combining demand prediction with adaptive inventory decision-making, the proposed framework aims to improve replenishment planning, reduce stockouts, minimize excess inventory, and support efficient inventory management. The system also provides inventory recommendations and analytical insights through a user-friendly dashboard. This integrated approach helps retailers make data-driven and timely inventory decisions.

Key Words :

Retail Inventory Management, Demand Forecasting, Machine Learning, Reinforcement Learning, Replenishment Planning, Predictive Analytics, Retail Analytics.